Overthinking Metaphor Visualization Tool - Product Requirements Document (Updated)

1. PROJECT OVERVIEW

Problem Statement: Based on user research with overthinkers and their supporters, three core problems emerged:

- **Expression Barrier:** "I don't know how to express how this feels" - difficulty articulating complex internal experiences
- **Validation Need:** "I want to feel seen, not fixed" - desire for understanding rather than clinical solutions
- **Communication Gap:** "I want others to get it without explaining everything" - struggle to help others understand their experience

Solution: A web application that translates emotional metaphors into interactive "data flowers" - radial visualizations that map algorithmic emotion analysis to petal lengths, creating a collective digital garden that makes emotional language patterns visible and comparable across overthinking experiences.

Primary Value Proposition: Bridges the gap between poetic metaphor and analytical understanding by providing a shared visual language for emotional expression patterns, helping observers and supporters recognize commonalities in how people describe complex internal experiences.

User Research Foundation:

- "It's hard for me to explain what's going on in my brain." —Olive
- "I feel like I'm behind glass." —Olivia
- "It's like feeding tangled yarn into a winder... that's what she tries to do when she writes... Writing is her primary method of communicating... I wish I could just show them." —Liyarui
- "Too much advice makes me spiral more." —Iris
- "I don't want to feel like I'm being told to fix something." —Chesley

Success Metrics:

- **MVP:** ≥90% of users successfully navigate garden views
- **MVP:** ≥40% QR scan rate from print zine to digital experience
- **MVP:** Garden loads within 1 second (73 flowers, lightweight)

- **MVP:** ≥70% of users explore multiple categories per session
- **MVP:** ≥4.2 minutes average session duration exploring emotional patterns

Research Validation Metrics:

- **Pattern Recognition:** Visual comparison enables recognition of similar emotional expressions across different metaphors
- **Analytical Understanding:** Algorithmic breakdown helps observers understand emotional language patterns
- **Empathy Building:** Comparative visualization helps supporters recognize complexity and variety in overthinking experiences

2. USER ANALYSIS & OBJECTIVES

Primary Users & Goals

Overthinkers (Primary)

- **Research-backed goals:** Find others with similar emotional patterns through reading authentic metaphors; see their own metaphorical expressions reflected in the collection; recognize they're not alone in their specific ways of describing internal experiences
- **Context:** People seeking validation through seeing similar emotional language patterns; looking for connection through shared metaphorical descriptions rather than advice
- **Success criteria:** Explore garden to find resonant metaphors; spend meaningful time reading others' expressions; feel understood through pattern recognition
- **Key insight:** Already use metaphorical language to express feelings ("tangled yarn," "behind glass"); benefit most from reading authentic experiences that mirror their own

Observers/Supporters (Secondary)

- **Research-backed goals:** Build empathy by understanding the variety and complexity of overthinking expressions; recognize patterns in emotional language; learn how people actually describe their experiences
- **Context:** Friends, family, or mental health supporters seeking insight into emotional language patterns and complexity
- **Success criteria:** Navigate garden meaningfully; understand emotional patterns through reading authentic metaphors; recognize the analytical complexity behind simple descriptions
- **Key insight:** Need tools that help them understand without offering unwanted advice ("Too much advice makes me spiral more" —Iris)

Researchers/Creators (Tertiary)

- **Goals:** Discover emotional patterns across metaphors; export insights for further analysis or creative work
- **Context:** Academic researchers, artists, or writers studying emotional expression
- **Success criteria:** Use filtering/sorting tools effectively; export meaningful data or visuals

Core User Journey

1. **Discovery:** Access via QR code from zine or direct web visit to Garden (main landing experience)
2. **Quick Insights:** Use shuffle feature at bottom of garden for curated random metaphor previews
3. **Exploration:** Browse 73-flower digital garden organized by overthinking categories that reflect interview themes ("feeling behind glass", "stuck in spirals") OR explore via shuffle preview cards
4. **Connection:** Click individual flowers or "Read More" to see detailed emotional breakdowns with enhanced modal content ("I wish I could just show them" —Liyarui)
5. **Reflection:** Browse similar metaphors in category views; share meaningful discoveries

3. CORE FEATURES & TECHNICAL SPECIFICATIONS

Phase 1: MVP Core Features

3.1 Garden (Main Landing & Exploration View)

Functional Requirements:

- Single-page landing experience combining welcome/orientation with full garden exploration
- Persistent minimal navigation bar with "Garden" (active/default) section
- Display all 73 pre-processed metaphors as interactive flowers using shared flower generation logic
- Pre-compute emotion analysis for all 73 survey responses during development phase (one-time processing)
- Store results in static JSON file for runtime access (no database required for MVP)
- Organize flowers by 7 overthinking categories in organic cluster layout (uneven distribution based on actual survey responses):
  - Loss of Agency (reflects "thoughts are running without me")
  - Thought Entanglement (reflects "feeding tangled yarn into a winder" —Liyarui)
  - Sensory Overwhelm (reflects experiences of static, numbness)
  - Perpetual Looping (reflects being "stuck in spirals")

    - Temporal Disconnection (reflects feeling disconnected from time/reality)
    - Perceptual Barriers (reflects "feeling behind glass" —Olivia, "in fog")
    - Emotional Dysregulation (reflects emotional stuckness, fragmentation)
- Enhanced flower detail modals with expanded "Read More" content
- Shuffle feature with preview cards at bottom of garden view for quick insights
- Smooth hover animations and idle "swaying" motion
- Category labels with actual flower counts (e.g., "Loss of Agency (12)" - reflecting real survey distribution)

Shuffle Feature Implementation:

- **Button Placement:** Fixed position at bottom of garden view
- **Preview Cards:** 3-card horizontal layout displaying random metaphor selections
- **Card Content:** Metaphor snippet (first 50 characters), category label, and "Read More" button
- **Randomization Logic:** Select 3 random metaphors from 73 responses, avoiding recent repeats
- **Session Tracking:** Track shown metaphors to prevent immediate duplicates
- **Integration:** "Read More" button opens same enhanced modal system as garden flowers
- **Refresh Capability:** "Shuffle" button regenerates 3 new random selections

3.2 Data Flower Visualization System

Shared Rendering System: Use central createFlower() function for all flower instances across the application

Technical Implementation:

- **Frontend:** Vanilla JavaScript with native SVG manipulation (no D3.js required)
- **Data Processing:** All emotion analysis already completed using Hugging Face Transformers - stored as static JSON
- **SVG Gradients & Textures:** SVG `<defs>` section with `<linearGradient>` and `<pattern>` elements for negative petals
- **Export:** Canvas-based PNG generation with designed "ID card" layout including flower visualization, metaphor text, category title, and emotion legend

SVG Coordinate System Requirements:

```javascript
```

```
// Required constants for accurate length mapping
const MAX_RADIUS = 150; // pixels from center to 100% ring
const LEGEND_RINGS = {
  ring25: MAX_RADIUS * 0.25, // 37.5px
  ring50: MAX_RADIUS * 0.50, // 75px
  ring75: MAX_RADIUS * 0.75, // 112.5px
  ring100: MAX_RADIUS // 150px
};

// Static emotion positioning (NEVER change this order)
const EMOTION_ANGLES = {
  // Negative zone: 0° to 120° (5 petals, 24° spacing each)
  fear: 12, anger: 36, disgust: 60, pessimism: 84, sadness: 108,
  // Neutral zone: 120° to 240° (2 petals, 60° spacing each)
  anticipation: 150, surprise: 210,
  // Positive zone: 240° to 360° (4 petals, 30° spacing each)
  optimism: 255, joy: 285, love: 315, trust: 345
};

// SVG Gradient and Texture Definitions
function createSVGDefs(svg) {
  const defs = document.createElementNS("http://www.w3.org/2000/svg", "defs");

  // SVG Gradient and Texture Definitions
function createSVGDefs(svg) {
  const defs = document.createElementNS("http://www.w3.org/2000/svg", "defs");

  // Gradient for negative petals
  const negativeGradient = document.createElementNS("http://www.w3.org/2000/svg", "linearGradient");
  negativeGradient.setAttribute("id", "negativeGradient");
  negativeGradient.innerHTML = `
    <stop offset="0%" style="stop-color:#005BAB;stop-opacity:1" />
    <stop offset="100%" style="stop-color:#4A90C2;stop-opacity:0.8" />
  `;

  // Gradient for neutral petals
  const neutralGradient = document.createElementNS("http://www.w3.org/2000/svg", "linearGradient");
  neutralGradient.setAttribute("id", "neutralGradient");
  neutralGradient.innerHTML = `
    <stop offset="0%" style="stop-color:#EEDE73;stop-opacity:1" />
    <stop offset="100%" style="stop-color:#F5F1A6;stop-opacity:0.8" />
  `;
```

```
  // Gradient for positive petals
  const positiveGradient = document.createElementNS("http://www.w3.org/2000/svg", "linearGradient");
  positiveGradient.setAttribute("id", "positiveGradient");
  positiveGradient.innerHTML = `
    <stop offset="0%" style="stop-color:#5EA748;stop-opacity:1" />
    <stop offset="100%" style="stop-color:#8BC474;stop-opacity:0.8" />
  `;

  // Texture pattern overlay (for negative petals only)
  const pattern = document.createElementNS("http://www.w3.org/2000/svg", "pattern");
  pattern.setAttribute("id", "textureOverlay");
  pattern.setAttribute("patternUnits", "userSpaceOnUse");
  pattern.setAttribute("width", "4");
  pattern.setAttribute("height", "4");
  // Add texture pattern content here

  defs.appendChild(negativeGradient);
  defs.appendChild(neutralGradient);
  defs.appendChild(positiveGradient);
  defs.appendChild(pattern);
  svg.appendChild(defs);
}

// Apply styling based on valence zone
function getVisualStyle(emotion) {
  const negativeEmotions = ['fear', 'anger', 'disgust', 'pessimism', 'sadness'];
  const neutralEmotions = ['anticipation', 'surprise'];
  const positiveEmotions = ['optimism', 'joy', 'love', 'trust'];

  if (negativeEmotions.includes(emotion)) {
    return {
      fill: "url(#negativeGradient)",
      filter: "url(#textureOverlay)"
    };
  } else if (neutralEmotions.includes(emotion)) {
    return {
      fill: "url(#neutralGradient)"
    };
  } else if (positiveEmotions.includes(emotion)) {
    return {
      fill: "url(#positiveGradient)"
    };
```

```
  }
}
```

Data Flower Visualization Specifications:

- **Length Mapping (Critical):** The radial length (distance from center to petal tip) must be a direct linear mapping of emotion percentage (0-100%)
  - Concentric legend rings define visual benchmarks:
    - 25% → 1st dotted ring from center
    - 50% → 2nd dotted ring from center
    - 75% → 3rd dotted ring from center
    - 100% → outermost edge of circle
  - Mathematical formula: `petalLength = (emotionPercentage / 100) × maxRadius`
  - Example: 83% emotion value must visually extend beyond the 75% ring to exactly 0.83 × maxRadius
- **Width:** Standard width for values ≥25%; thinner width for <25% (prevents center blob)
- **Shape:** Smooth, tapered petals radiating from center point (no visible hub)
- **Dominant Emotion:** Solid fill for highest-scoring emotion
- **Other Emotions:** Outline-only rendering
- **Animation:** Bloom effect from center outward over 1.2 seconds

Emotion Mapping & Petal Structure:

- **Total:** 11 petals in consistent clockwise order
- **Valence Zones (equal ⅓ divisions):**
  - **Negative (Petals 1-5):** Blue gradient (`#005BAB` to lighter blue) with texture overlay
    1. Fear, 2. Anger, 3. Disgust, 4. Pessimism, 5. Sadness
  - **Neutral (Petals 6-7):** Yellow gradient (`#EEDE73` to lighter yellow) 6. Anticipation, 7. Surprise
  - **Positive (Petals 8-11):** Green gradient (`#5EA748` to lighter green) 8. Optimism, 9. Joy, 10. Love, 11. Trust

Interaction Specifications:

- **Individual Petal Hover:** Display tooltip with format "[Emotion]: [Percentage]%" (e.g., "Sadness: 72%")

- **Zone Label Hover:** Highlight all petals within valence zone and show valence proportion calculation
- **Valence Formula:** (Sum of zone emotion intensities) ÷ (Sum of ALL emotion intensities) × 100
- **Example:** (179% negative emotions) ÷ (247% total emotions) × 100 = "Negative emotions: 72% of this experience"
- **Visual:** All petals in zone glow/brighten while others dim slightly

3.3 Enhanced Modal Content with Overlay Layout

Modal Structure:

- **Background:** Preserves original screen (category view or garden) with semi-transparent cloudy overlay
- **Content Layout:** Responsive design with flower visualization and metadata display

Desktop Layout (Horizontal):

- **Left side (50% width):** Interactive flower visualization with hover states preserved
- **Right side (50% width):** Vertically stacked information hierarchy
    - Category header (Junicode Regular, 10% letter spacing, #1A2017)
    - Full metaphor text in quotes (Satoshi Variable Regular, body text, #1A2017)
    - Emotional Intensity: "[percentage]%" with label
    - Dominant Valence: "Positive/Negative/Neutral" with label
    - Dominant Emotion: Specific emotion name with label

Mobile Layout (Vertical):

- **Top section:** Interactive flower visualization
- **Bottom section:** Two-column layout below flower
    - **Left column (60% width):** Category header (Junicode Regular, 10% letter spacing) and full metaphor text in quotes
    - **Right column (40% width):** Data metrics stacked vertically
        - Emotional Intensity: "[percentage]%" with label
        - Dominant Valence: "Positive/Negative/Neutral" with label
        - Dominant Emotion: Specific emotion name with label

Visual Treatment:

- **Overlay Effect:** Semi-transparent cloudy background that dims underlying content
- **Modal Container:** Clean white/light background with rounded corners
- **Information Labels:** Small descriptive text above each data point
- **Flower Integration:** Same interactive flower with hover tooltips preserved

Interaction Specifications:

- **Modal Trigger:** Click any flower from category pages or shuffle cards
- **Modal Close:** ESC key, outside click, or "Back" button
- **Flower Interactivity:** Maintains petal hover states showing emotion percentages
- **Responsive Breakpoints:** Switches from horizontal to vertical layout at tablet breakpoint

Modal Specifications:

- **Modal Trigger:** Click flower opens enhanced detailed overlay with rounded corners (border-radius: 12px)
- **Modal Close:** ESC key, outside click, or X button (top-right corner)
- **Navigation Access:** Top navigation bar with "Garden" section

3.4 Flower Positioning System

Static Position Mapping: Use hardcoded coordinates for consistent, artistically controlled layouts

```javascript
```

```
// positions.js
const categoryLayouts = {
  "Sensory Overwhelm": [
    { id: 12, x: 200, y: 300 }, // flower ID 12 at these coordinates
    { id: 45, x: 800, y: 150 }, // flower ID 45 at these coordinates
    { id: 7, x: 400, y: 500 },  // etc.
  ],
  "Loss of Agency": [
    // different positions for this category
  ]
  // ... other categories
};

// Render flowers at designated positions
function renderCategoryPage(categoryName) {
  const positions = categoryLayouts[categoryName];

  positions.forEach(pos => {
    const flowerData = metaphorDatabase.find(f => f.id === pos.id);
    const container = createPositionedContainer(pos.x, pos.y);
    createFlower(flowerData, container.id);
  });
}
```

Layout Specifications:

- **Garden Style:** Organic clustering with subtle randomization within category bounds
- **Flower Scaling:** Base size + scaling factor based on total emotional intensity
- **Category Grouping:** Visual separation between categories using whitespace and subtle background color shifts (7 clusters with natural survey distribution)
- **Responsive Design:** CSS Grid with flexible flower positioning for mobile/desktop
- **Animation:** Gentle swaying motion using CSS transforms (respects prefers-reduced-motion)
- **Performance:** Standard loading (73 flowers is lightweight, no lazy loading needed)

3.5 Dataset Strategy & Structure

Garden Display: Only the original 73 survey responses are shown to users

- Authentic, real experiences from actual respondents
- Uneven category distribution (reflects natural survey results)

- Each flower shows genuine overthinking metaphors for connection/validation

```javascript
// Data structure example
{
  "gardenMetaphors": [
    {
      "id": 1,
      "text": "I feel like my thoughts are running without me",
      "category": "Loss of Agency",
      "source": "survey",
      "emotions": {
        "fear": 0.13, "anger": 0.06, "disgust": 0.26,
        "pessimism": 0.51, "sadness": 0.83, "anticipation": 0.14,
        "surprise": 0.01, "optimism": 0.21, "joy": 0.02,
        "love": 0.00, "trust": 0.01
      },
      "dominantEmotion": "sadness",
      "totalIntensity": 2.47
    }
    // ... all 73 survey responses
  ],
  "categories": [
    "Loss of Agency",
    "Thought Entanglement",
    "Sensory Overwhelm",
    "Perpetual Looping",
    "Temporal Disconnection",
    "Perceptual Barriers",
    "Emotional Dysregulation"
  ]
}
```

Implementation Strategy:

```javascript
```

```
// Shared flower generation function
function createFlower(metaphorData, containerId) {
  // Same SVG logic for all flowers
  // metaphorData.emotions contains the pre-computed scores
  // Renders petals, applies hover handlers, etc.
}

// Garden initialization
function initializeGarden() {
  fetch('/garden-data.json')
    .then(response => response.json())
    .then(data => {
      data.metaphors.forEach(metaphor => {
        createFlower(metaphor, `garden-flower-${metaphor.id}`);
      });
    });
}
```

4. TECHNICAL ARCHITECTURE

4.1 Frontend Stack

Core Technologies:

- HTML5/CSS3/Vanilla JavaScript (no framework overhead)
- Native SVG manipulation for data visualization and animation (no D3.js required)
- CSS Grid + Flexbox for responsive layouts
- Canvas API for PNG export functionality

Typography Requirements:

- Junicode (Google Fonts or self-hosted)
- Satoshi Variable (self-hosted via font files)
- Font loading optimization with font-display: swap
- Fallback fonts: serif for Junicode, sans-serif for Satoshi

Performance Optimizations:

- SVG sprite sheets for repeated flower elements
- RequestAnimationFrame for smooth animations
- Intersection Observer for garden flower lazy loading

- CSS containment for modal overlays

4.2 Backend Architecture

Server Environment:

- Node.js/Express for API endpoints (optional - can be fully static)
- Static JSON file storage for 73 metaphors with pre-computed emotion scores

API Endpoints:

- GET /garden-data (→ 73 metaphors JSON)
- POST /export-flower (SVG → PNG conversion)

4.3 Hosting & Deployment

Recommended Stack:

- Frontend: Netlify/Vercel (free tier, CDN)
- Backend: Railway ($5/month) or Render (free tier with limitations)
- Alternative: Hugging Face Spaces (free ML model hosting)

Domain & SSL:

- Custom domain via hosting provider
- Automatic HTTPS via Let's Encrypt

5. ACCESSIBILITY & DESIGN REQUIREMENTS

5.1 Non-Negotiable Accessibility Features

- **Color Accessibility:** High contrast ratios, colorblind-safe palette variations
- **Keyboard Navigation:** Full tab navigation, ESC modal closing, focus indicators
- **Screen Reader Support:** ARIA labels for all flower elements, semantic HTML structure
- **Motion Control:** Respects prefers-reduced-motion, toggle for animations
- **Sound Control:** No auto-playing audio, user-controlled sound toggle

5.2 Design Principles

- **Pattern Recognition over Experience Simulation:** Visual choices that enable comparison and analysis of emotional language patterns, not replication of subjective feelings

- **Analytical Understanding over Clinical Diagnosis:** Provides algorithmic breakdowns of emotional expression to build empathy and recognition, without suggesting medical interpretation
- **Comparative Exploration over Individual Expression:** Enables users to see how their emotional language relates to others' metaphorical descriptions
- **Educational Interaction:** Clear, informative feedback that helps users understand emotional language analysis without being prescriptive
- **Accessible Complexity:** Makes sophisticated emotional analysis approachable through visual design

5.3 Visual Hierarchy

Typography System:

- **Title Screen Headers:** Junicode Italic, large size, white text on blue background
  - Usage: Main introductory screens, hero statements
  - Example: "What does overthinking feel like?" (Image 3)
- **Navigation & Interface Labels:** Satoshi Variable, ALL CAPS, smaller size
  - Usage: Button text, section headers, interface controls
  - Example: "EXPLORE BY THEME", "SHUFFLE THOUGHTS", "BACK"
- **Category Names:** Junicode Regular, 10% letter spacing, #1A2017
  - Usage: Category labels throughout garden and category pages
  - Example: "Sensory Overwhelm", "Perpetual Looping", "Loss of Agency"
- **Modal Subheaders:** Junicode Regular, 10% letter spacing, #1A2017
  - Usage: Data labels in flower detail modals
  - Example: "Emotional Intensity", "Dominant Valence", "Dominant Emotion"
- **Body Text & Descriptions:** Satoshi Variable Regular, #1A2017, 16px minimum
  - Usage: Metaphor quotes, reflection snippets, descriptive text
  - Line height: 1.5 for optimal readability
  - Example: "A heavy, buzzing cloud over my head throughout the day"

Color Palette:

- Primary: Blue #005BAB, Yellow #EEDE73, Green #5EA748
- Text: Dark #1A2017 for high contrast readability
- Background: Soft neutral (#FAFAFA or #F8F9FA)
- Accents: Subtle gradients and soft shadows for depth

6. DEVELOPMENT PHASES & DELIVERABLES

Phase 1: MVP (Week 1-3)

Core Deliverables:

- 73-flower digital garden with category organization
- Shared flower generation system using vanilla JavaScript + SVG
- Modal system for detailed flower views
- PNG export functionality
- Basic responsive design
- Essential accessibility features

Success Criteria:

- Functional garden browsing with smooth interactions
- Export works across major browsers
- Meets WCAG 2.1 AA for critical features
- Page load under 1 second

Phase 2: Enhancement (Week 4-6)

Additional Features:

- Enhanced accessibility features
- Performance optimizations
- Refined animations and micro-interactions

Success Criteria:

- Smooth view transitions
- Maintains performance with full feature set

7. RISK MITIGATION & ALTERNATIVES

7.1 Technical Risks

- **Flower Rendering Performance:** Use efficient SVG generation, avoid complex animations on slower devices
- **Browser Compatibility:** Progressive enhancement ensures basic functionality across browsers

- **Hosting Costs:** Start with free tiers, upgrade incrementally based on usage

7.2 User Experience Risks

- **Complexity Overwhelm:** Provide clear onboarding and contextual help
- **Emotional Sensitivity:** Include gentle disclaimers about emotional content
- **Accessibility Gaps:** Regular testing with assistive technologies

This PRD provides clear technical specifications while preserving the artistic vision and ensuring actionable development steps focused on viewing and exploration rather than creation.